

THE STRUCTURE OF MATHEMATICAL REALITY

Phase I — Chapter 3

The Map of Mathematical Fields

Chapter 3 — Phase I: What Mathematics Is

Self-Study Curriculum

What This Map Is For

- Not to file ideas into boxes.
- To locate yourself when your own work stalls.
- Name the field you are standing in, find the tool you need.

A good map says what each region is *for* and how to cross between them.

Each Field, One Central Question

Field	It asks
Algebra	what are the symmetries of this structure?
Analysis	what happens in the limit?
Geometry	what survives rigid motion?
Topology	what survives continuous deformation?
Number theory	how are the primes distributed?
Logic	what can (and cannot) be proved?
Probability	what can be inferred under uncertainty?

Orientations, Not Definitions

- The question is a *compass bearing*, telling you which way a field faces.
- One object can be interrogated by several fields at once.
- Each field's question extracts something different from it.
- That overlap is the map's most useful feature, not a defect.

The Borders Hold the Power

When two fields ask the same question about different objects, a bridge forms.

- A bridge beats either field alone: carry a hard question across, answer, carry back.
- Inside one field you are stuck with one toolkit; a bridge frees you.
- The border is not a boundary. It is a translation.

Four Load-Bearing Bridges

- **Algebraic geometry:** polynomial equations = geometric shapes.
- **Analytic number theory:** continuous functions govern the discrete primes.
- **Differential geometry:** calculus on curved surfaces (and curved data).
- **Algebraic topology:** attach algebra to shapes to tell them apart.
- Each lets one field's tools answer another field's question.

Bridges Already in Your Work

- Persistent homology (TDA) = algebraic topology on point clouds.
- The manifold hypothesis = differential geometry on high-dimensional data.
- Steady states of a reaction network = algebraic geometry (polynomial solution sets).
- Robust features survive noise because they are *topological*, not metric.

One Object, Three Fields: The Eigenvalue

- Algebraic: a root of the characteristic polynomial.
- Analytic: sets the rates and long-term behavior of a system.
- Geometric: a stretch factor along a principal axis.
- Biologist, physicist, geometer are not disagreeing; each names one facet.
- Which field you adopt decides which powers you may use.

Same Decomposition, Two Readings

- PCA: eigenvalues *geometrically* = variance along principal axes.
- Markov chain: eigenvalues *analytically* = stationary state and mixing rate.
- The math object is identical; the question you bring changes the meaning.
- Eigendecomposition is powerful because it works *along the grain* (Ch 5, Ch 8).

“Are These Two Clusters Different?”

- Probability: unlikely to share one underlying distribution (a test).
- Geometry: centers far apart relative to spread (metric distance).
- Topology: two disconnected pieces, not one blob (connectivity).
- Same question, three fields, three non-interchangeable meanings.
- Knowing which you want is half of asking well.

Hierarchy: How Much Structure Is Kept

- Geometry (distance, angle) $\rightarrow$ forget distance $\rightarrow$
- Topology (only continuity) $\rightarrow$ forget continuity $\rightarrow$
- Set theory (only membership).
- Each step forgets something; what remains is more general, less powerful.
- More structure = fewer situations, stronger results; less = more situations, weaker.

The Algebraic Ladder

- Fields ($+$, $\times$, both invertible: $\mathbb{R}$, $\mathbb{C}$) $\rightarrow$ forget $\times$ inverses $\rightarrow$
- Rings (two operations, $\times$ not invertible: $\mathbb{Z}$) $\rightarrow$ drop an operation $\rightarrow$
- Groups (one operation, with inverses) $\rightarrow$ forget inverses $\rightarrow$
- Monoids (one associative operation: $\mathbb{N}$ under $+$).
- A theorem proved low on the ladder holds everywhere above it (Ch 7).

Cube = Sphere, Donut Different

- Topology already forgot distance, angle, curvature.
- What it keeps is connectivity and holes.
- Euler characteristic: cube 2, sphere 2, donut 0.
- “Topologically protected” = survives all continuous deformation.
- Only a tear or glue (a discontinuous event) can destroy it, never mere noise.

The Many Meanings of Distance

- Euclidean distance: geometry, most specific, richest (trigonometry).
- Metric (three axioms): analysis, holds for any sensible space.
- Neighborhoods: topology, only “near,” no “how near.”
- KL divergence: information theory, not even a metric (asymmetric).

Most structure and least general, down to least structure and most general.

Choosing the Right Generality

- Too specific: your result is a one-off, true of your data and nothing else.
- Too general: nothing strong is left to say.
- The skill: use the *least* structure the problem actually needs.
- Then your conclusion is as portable as the problem allows.
- You feel this every time you weigh a bespoke model against a general method.

Where You Now Stand

- Fields are oriented by a question, joined by bridges, ranked by structure kept.
- One object can live at several fields at once; pick the field, unlock its powers.
- Choosing generality is now a navigable axis, not a matter of taste.

One cell waited: Logic, “what can be proved?” Chapter 4 opens Phase II, formal systems, proof as mechanism, and Gödel’s limit where Chapter 2’s diagonal returns.